8

Motions of the Earth

Topics covered

- The earth's axis
- Rotation causes day and night
- Revolution
- Leap year
- How the seasons happen
- The rhythms of the earth's movements

The earth spins on its axis. Do this experiment to understand what an axis is. You need a bangle and a table tennis ball.

- Spin the bangle. You will see a central line around which the bangle spins. This is its axis. The axis does not exist—it is an imaginary line.
- Label the axis in this picture of a spinning bangle.
- Now spin the ball. It will also spin around an axis, but you will not be able to see it.

A spinning bangle

THE EARTH'S AXIS

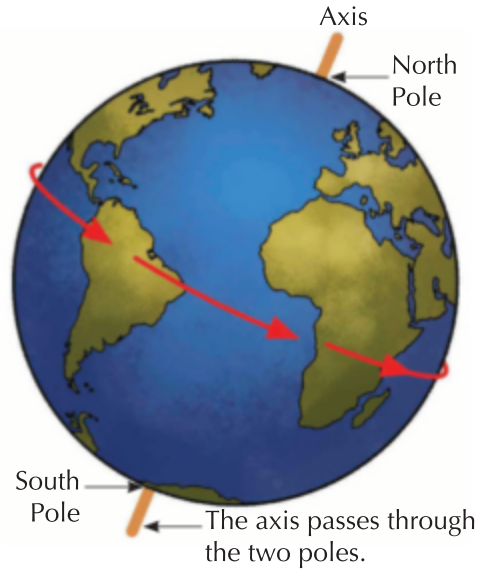
Like the ball, the earth also spins, or rotates, around its axis. We cannot see this axis. Look at a globe, which is a model of the earth. It will help you to understand some more things about the earth.

- The axis of the globe is **tilted** (slanting)—it is not **vertical**. The axis of the earth is also tilted in the same way.
- The axis of the globe passes through the top and the bottom of the globe. In the same way the axis of the earth passes through the north and the south of the earth. In the north, this point is called the North Pole. In the south, it is called the South Pole.

- The earth always spins in the same direction around its axis—from west to east. Spin the globe to see how this happens.

The diagram shows you the axis and the two poles of the earth. You can also see the tilted axis of the earth.

The spinning movement of the earth is called rotation.

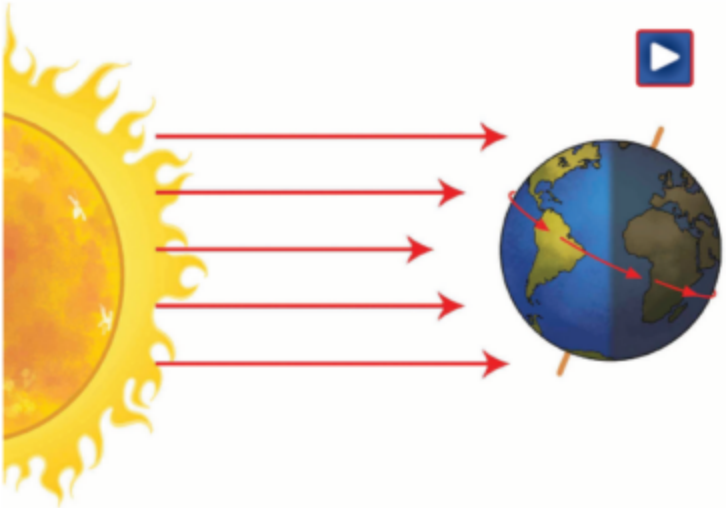


The arrows show the direction in which the earth spins—from west to east.

ROTATION CAUSES DAY AND NIGHT

The earth takes 24 hours to complete one rotation. This is what happens during a rotation:

- As the earth rotates, the parts of the earth facing the sun get light. These areas have day. (See diagram).
- The opposite side of the earth, which does not get the sun's light, has night.
- During the 24 hours that the earth takes for one rotation, every part of the earth will have day and night.



Day and night

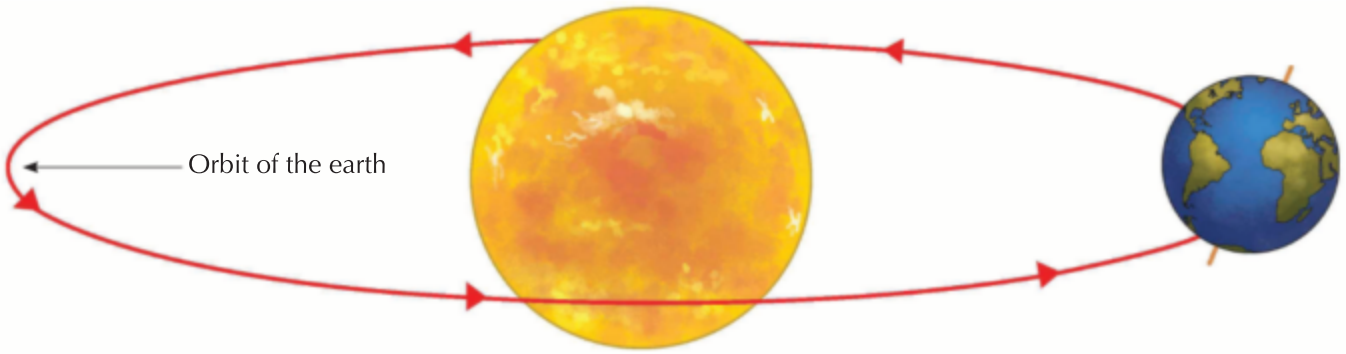
One complete day on earth is 24 hours long because the earth takes 24 hours to complete one rotation. The earth spins continuously so we have a continuous cycle of day and night.

Think and answer

- If the earth did not rotate on its axis, how would this affect the earth (hint: heat, dark)?
- Why is the earth's rotation important for us?

REVOLUTION

The earth does not seem to move, but it is moving all the time. It rotates on its axis. At the same time it also moves around the sun in its orbit. This movement is called revolution. It takes 365 days and 6 hours to complete one revolution around the sun. This is why a year usually has 365 days.



The earth revolves around the sun.

LEAP YEAR

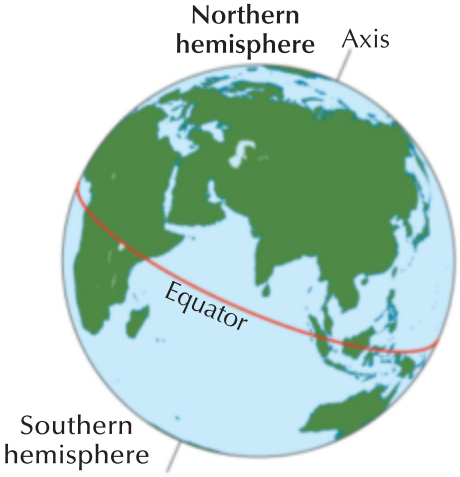
- A year usually has 365 days, but the earth takes 365 days and 6 hours for a complete orbit.
- The 6 extra hours from each year are added up. In four years these become 24 hours (6x4=24). These 24 hours are equal to one day. This is an extra day.
- Every fourth year, this extra day is added to the year. The year with an extra day is called a **leap year**. The extra day is added to February, so in a leap year February has 29 days.
- A leap year has 366 days and it comes every 4 years—2008, 2012 and 2016 were all leap years.

HOW THE SEASONS HAPPEN

The change in seasons from summer to winter and then back from winter to summer happens because of two reasons:

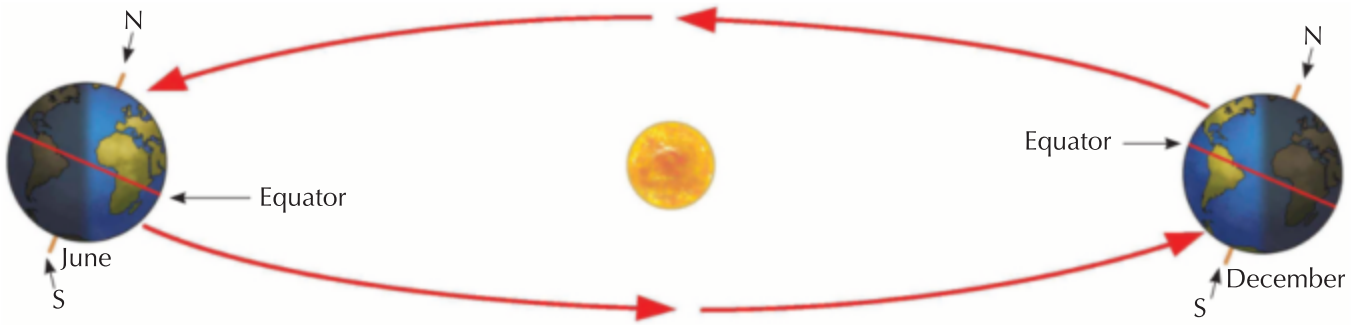
- the tilt of the earth's axis
- the orbit of the earth around the sun

The diagram shows the tilt of the earth's axis. It also shows that the equator divides the earth into two equal halves. These are the northern **hemisphere** and the southern hemisphere (hemi=half, sphere=a round shape). India is in the northern hemisphere.



The earth's hemispheres

This drawing shows the orbit of the earth. It shows the earth in two positions on its orbit—in June and in December. Let us see what happens when the earth is in these two positions.



The orbit of the earth around the sun

In June

TP
SI

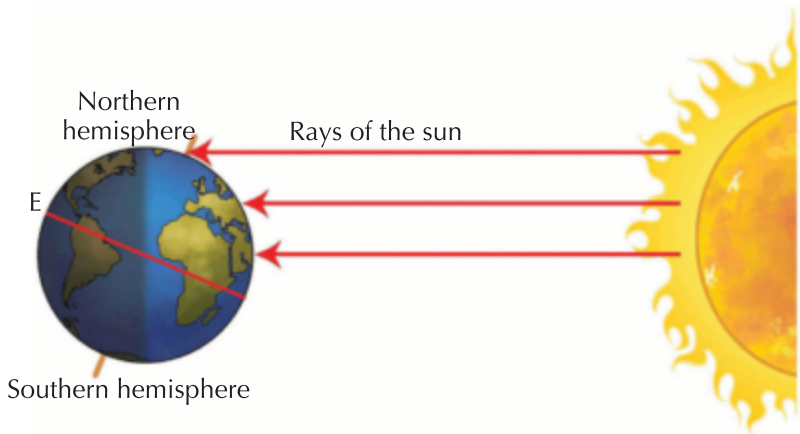
You will need a big globe and an adjustable reading lamp.

1. Place the lamp at the centre of a table; it represents the sun.
2. Look at the diagram of the earth's orbit and place the globe in the position that says 'June'. Make sure that the axis is tilted in the same direction as in the diagram.
3. Turn on the lamp and focus the light on the globe.

Understand the following paragraphs with the help of the globe and the lamp.

The northern hemisphere has *summer in June*. This is what happens:

- The northern hemisphere is tilted *towards* the sun and the rays of the sun fall *directly* on it. Look at your globe to see this.



The northern and southern hemispheres in June

- The days are long because the sun rises early and sets late.
- Look at the North Pole. As the earth spins, the sun never sets at the North Pole in June. So, the North Pole and the areas around it will have continuous day and never have night in June! This is called **polar day,** and the sun is called the midnight sun. Spin the globe to see how this happens.

The southern hemisphere has *winter in June*. Here is how:

- The southern hemisphere is tilted *away* from the sun. The rays of the sun fall *less directly* on this hemisphere. Look at your globe to understand this.
- The days are short because the sun rises late and sets early.
- Look at the South Pole. As the earth rotates, the sun never rises here. The South Pole and its surrounding areas are always dark in June! This is called polar night. If you spin the globe, you will understand this.

So, in June, while the northern hemisphere has summer, the southern hemisphere has winter.

In December

TP

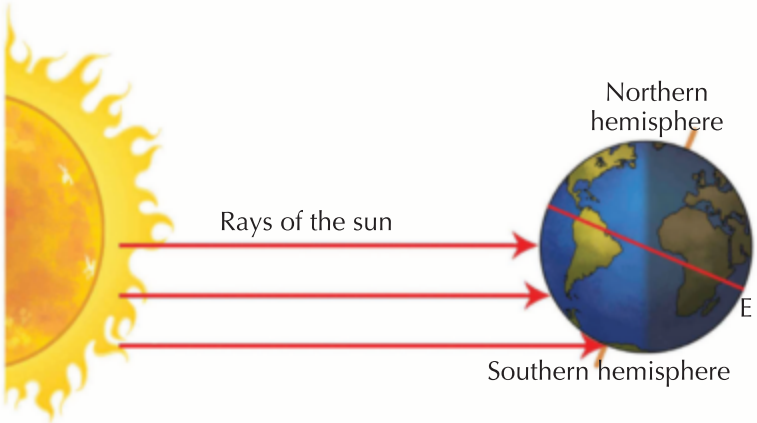
SI

1. Now shift your globe to the December position in its orbit. This is the opposite of the June position. Make sure the axis is tilted in the same direction as the diagram.
2. Turn the lamp to face the globe.

In December the seasons will be the opposite of what they are in June.

The northern hemisphere has *winter in December*:

- The northern hemisphere is tilted *away* from the sun. The sun's rays fall *less directly* on it.
- The days are short as the sun rises late and sets early.



The northern and southern hemispheres in December

- The North Pole will now have continuous night—polar night. Look at your globe, and spin it, to understand this.

The southern hemisphere has *summer in December*:

- The southern hemisphere is tilted *towards* the sun. Its rays will fall *directly* on the hemisphere.
- The days are long as the sun rises early and sets late.
- The South Pole will have continuous day. Use your globe to understand this.

Did you Know?

The northern hemisphere has the longest day on or about June 21 and the shortest day on or about December 22. Day and night are equal on or about March 21 and September 22.

The two hemispheres have *opposite seasons*. When the northern hemisphere (in places like India, the USA or Europe) has summer, the southern hemisphere (in places like Australia and South Africa) has winter. When the southern hemisphere has summer, it is winter in the northern hemisphere.

Did you Know?

Six seasons were identified in ancient India with the following Sanskrit names: *Grishm* (summer), *Varsha* (monsoon), *Sharad* (autumn), *Hemant* (winter), *Shishir* (the season when snow thaws), and *Vasant* (spring).

THE RHYTHMS OF THE EARTH'S MOVEMENTS

We have seasons because of the tilt of the earth's axis and the orbit of the earth around the sun. As the earth goes around the sun, one season follows another in a continuous cycle. This happens year after year.

The **rhythm** (regular movement) of the seasons never changes because the revolution of the earth never changes.

In the same way, the rhythm of day and night remains the same because the earth always takes 24 hours for one rotation.

Discuss

- How do the long nights of winter affect you?
- What effect do the long days of summer have on you?

The rhythms of the earth's movements are important for us because our lives are affected by the seasons and by the daily pattern of day and night.

Values and Life Skills

- How should we adjust our routine according to the rhythms of the earth's movements?
- Why is this important?

REMEMBER

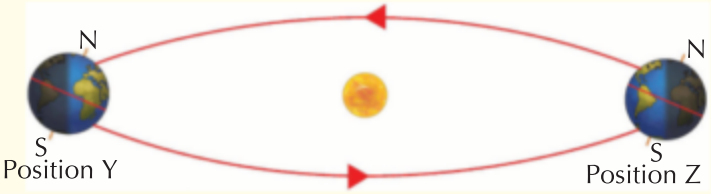
1. The axis of the earth is an imaginary line passing through the two poles; it is slightly tilted.
2. The earth has two types of movement, rotation and revolution.
3. The earth's rotation takes 24 hours and causes day and night.
4. The earth's revolution takes 365 days and 6 hours.
5. The extra 6 hours add up to give an extra day every 4 years. This is a leap year which has 366 days.
6. The summer and winter seasons are caused by (a) the revolution of the earth and (b) the tilt of the earth's axis.
7. The equator divides the earth into two equal hemispheres.
8. The seasons are always opposite in the two hemispheres.
9. The rhythms of the earth's movements never change so the rhythm of the seasons and of day and night never changes.

NEW WORDS

Tilted	sloping; this is a tilted line: /
Vertical	absolutely straight; this is a vertical line: |
Hemisphere	half a sphere
Rhythm	regular movement

EXERCISES

A. Fill in the blanks.



1. Fill the blanks in these sentences with the help of the diagram.
   a. In position Y, it is the ..... season in the northern hemisphere and the ..... season in the southern hemisphere.
   b. In position Z, it is the ..... season in the southern hemisphere and the ..... season in the northern hemisphere.
2. Next to each position in the diagram, write J or D to show which month it is.

B. Put a circle around the odd one out.

In your notebook, explain why it is not a part of the list.

1. rotation, spin, earring, bangle, ball
2. 29 February, orbit, leap year, 366, sun's rays
3. day and night, spin, dizzy, axis, 24 hours
4. one year, revolution, 365 days 6 hours, celebration, orbit
5. winter, polar night, polar bear, hemisphere tiled away from the sun

C. Write true or false.

Correct the false sentences in your notebook.

1. The axis is an imaginary line running from north to south through the centre of the earth.
2. The earth's axis is tilted slightly.
3. The earth rotates from east to west.
4. The earth completes one revolution in 366 days.
5. The seasons are opposite in the two hemispheres.
6. The rhythms of the earth's movements never change.

D. Choose the correct answer.

1. One reason for the seasons is the **rotation**/**revolution** of the earth.
2. The sun rises early and sets late in **summer**/**winter**.
3. The sun rises late and sets early in **summer**/**winter**.
4. In winter the polar regions have **continuous night**/**day and night**.
5. In winter the sun's rays are **more direct**/**less direct**.

E. Answer the following questions.

1. What is the axis of the earth?
2. How long does the earth take for one rotation?
3. What does the rotation of the earth cause?

4. What is the earth's revolution?
5. How long does the earth take for one revolution?
6. Write a short paragraph to explain why a leap year has 366 days.
7. Give the two reasons why summer and winter happen.
8. Why is the northern hemisphere (a) hotter than the southern hemisphere during summer and (b) colder than the southern hemisphere during winter?
9. What happens in the North Pole during summer?

THINGS TO DO

A. Picture study

This is a picture of the earth taken from space. Discuss:

1. Why is the full sphere of the earth not visible?
2. Which side of the earth is facing the sun?

B. On your own

Do some earth maths!

The earth moves extremely fast in its revolution around the sun—at a speed of 30 kilometres per second which is 1,800 kilometres in one minute! How much will it travel during your class period of 30 or 35 minutes?

What is the speed per hour of the vehicle that you travel in when you go to school?

TP C. In a group

1. Have some fun with a globe.

   You will need: • A powerful torch • A globe • A darkened room

   a. In a darkened room, ask a friend to hold the lit torch pointing towards the globe.
   b. Meanwhile, hold the globe and rotate it from west to east.
   c. Which country in Asia gets the light from the torch first? Which states of India get the light first?

2. Act it out!

   You and your friends can do plays about the sun and the earth. Make four or five groups. Each group should:

   - Write a play with two characters—the sun and the earth. One person can be the storyteller. Add other characters, if you wish.
   - Decide the costumes and give a name to the play. Practise and perform it for your classmates and other classes to see.

WEBLINK

Earth's Revolution & Rotation: https://www.youtube.com/watch?v=EXasopxAFoM (accessed on 16 Aug 2017)